

Assignment 3

Day5 Case Study practical(1 hour): Work on a scenario involving multiple hardware issues.

Task 1: Laptop Not Powering On

Scenario

A user reports that their laptop does not power on even when it is connected to the charger.

Troubleshooting Steps

Step 1 – Check the power socket

First, check whether the wall socket is providing power. A different working device can be connected to the same socket for testing.

What I am checking:

Whether the problem is caused by the electrical power source.

Step 2 – Check the charger and cable

Inspect the laptop charger, adapter and cable for cuts, loose connections, burning marks or other physical damage.

What I am checking:

Whether the charger is supplying power to the laptop.

Step 3 – Check the charging port

Inspect the laptop's charging port for dust, damage or a loose connection.

What I am checking:

Whether the charger is making proper contact with the laptop.

Step 4 – Check the battery

If the battery is removable/accessibly serviceable, check its connection and condition.

What I am checking:

Whether a battery problem is preventing the laptop from starting.

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Step 5 – Perform a power reset

Disconnect the charger. If the battery is removable, remove it. Hold the power button for about **15–30 seconds**, reconnect the appropriate power source and try starting the laptop.

What I am checking:

Whether a temporary power-state problem is preventing startup.

Conclusion : If the laptop still does not power on after these checks, the fault may be related to the **charger, battery, DC charging circuit, motherboard or another internal component.**



Fig 1: Laptop charger and power connection
(checking the power source and charger)



Fig 2: Checking the laptop's
charging port

Task 2: Monitor Shows a Blank Screen

Troubleshooting Flow

Step 1 – Check monitor power

Make sure the monitor is switched ON and its power cable is properly connected.

Step 2 – Check the display cable

Check the HDMI, DisplayPort or VGA cable at both the monitor and computer ends.

Step 3 – Try another cable or port

If available, test with another working display cable or another video port.

Step 4 – Check monitor input source

Make sure the monitor is set to the correct input, such as **HDMI, DisplayPort or VGA**.

Step 5 – Check the graphics card

For a desktop with a dedicated GPU, check that the graphics card is properly seated and its required power connectors are attached.

Step 6 – Test with another monitor

Connect a known-working monitor to determine whether the problem is with the original monitor or the computer.

Step 7 – Check display settings

If the system can display through another monitor, check the Windows display settings and graphics driver.

Conclusion: The blank-screen problem can be caused by a loose display cable, incorrect monitor input, faulty monitor, GPU connection problem or display-driver/settings issue.



Fig 3: Blank screen on monitor
(typical issue)



Fig 4: Checking GPU connection
and power cables



Fig 5: Monitor input source
settings

Task 3: Wireless Mouse Not Responding

Scenario

The wireless mouse is not responding, but the keyboard is working normally.

Troubleshooting Procedure

Step 1 – Check the mouse battery

Replace or recharge the mouse battery and check whether the mouse starts working.

What I am checking:

Whether the problem is caused by a discharged battery.

Step 2 – Check the wireless receiver

Remove the USB receiver and connect it to another USB port.

What I am checking:

Whether the receiver or original USB port is causing the problem.

Step 3 – Test another USB port

If the mouse works in another port, the original USB port may have a problem.

Step 4 – Test another mouse

Connect another known-working wireless/wired mouse to the same system.

What I am checking:

If another mouse works, the original mouse or receiver may be faulty.

Step 5 – Check system settings

Open **Device Manager** → **Mice and other pointing devices** and check whether the mouse is detected correctly.

Step 6 – Check/reinstall the driver

If necessary, update or reinstall the appropriate mouse/USB device driver.



Fig 6: Wireless mouse and USB receiver (check if mouse works)

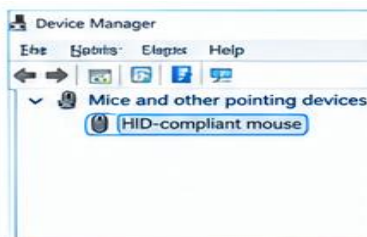


Fig 7: Device Manager (check if mouse is enabled)



Fig 8: Try a different USB port (to rule out port issue)

Task 4: Gaming PC Overheating and Shutting Down

Scenario

A gaming-cafe PC becomes very hot during use and suddenly shuts down.

Three Possible Hardware Causes

1. CPU Cooler/Fan Problem

How to check:

- Open the PC case after shutting down and disconnecting power.
- Check whether the CPU cooling fan is spinning properly.
- Check for unusual fan noise.
- Check whether the heatsink is properly attached.
- Clean the fan if it is blocked by dust.

Possible problem

A failed or blocked CPU fan can cause the CPU temperature to rise and may result in thermal shutdown.

2. Dust Build-up

How to check:

- Inspect the CPU fan, heatsink and case vents.
- Look for accumulated dust.
- Check whether airflow is blocked.
- Clean the dust using appropriate cleaning equipment and safe procedures.

Possible problem

Heavy dust can restrict airflow and reduce cooling efficiency, causing the system to overheat.

3. Thermal Paste Problem

How to check:

- If trained and authorized to do so, inspect the thermal interface between the CPU and cooler.
- Old or improperly applied thermal paste may reduce heat transfer.
- Reapply suitable thermal paste correctly when required.



Fig 9: Checking CPU fan
(should be spinning)



Fig 10: Dust build-up
(on fan and heatsink)



Fig 11: Thermal paste
(check for dryness)